PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME						
CT GROUP	1	7	S			

INDEX NUMBER

BIOLOGY 9744/04

Paper 4 Practical 13 Aug 2018

2 hours 30 minutes

Candidates answer on the Question Paper
Additional materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your CT group, index number and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	24
2	17
3	14
Total	55

This document consists of **17** printed pages and **1** blank page.

[Turn over

1 The enzyme amylase, **E**, hydrolyses (breaks down) starch, to a reducing sugar.
You are required to investigate how much reducing sugar diffuses from a mixture of starch and amylase through a partially permeable wall of Visking tubing.

You are provided with:

labelled	contents	hazard	volume /cm^3
E	2.0% amylase solution	irritant	20
S	1.0% starch suspension	none	20
W	distilled water	none	250

labelled	contents	hazard	details	quantity
V	Visking tubing	none	15 cm length in distilled water	1

If **E** comes into contact with your skin, wash it off immediately under cold water.

It is recommended that you wear suitable eye protection.

Fig. 1.1 shows the apparatus you will set up for this investigation.

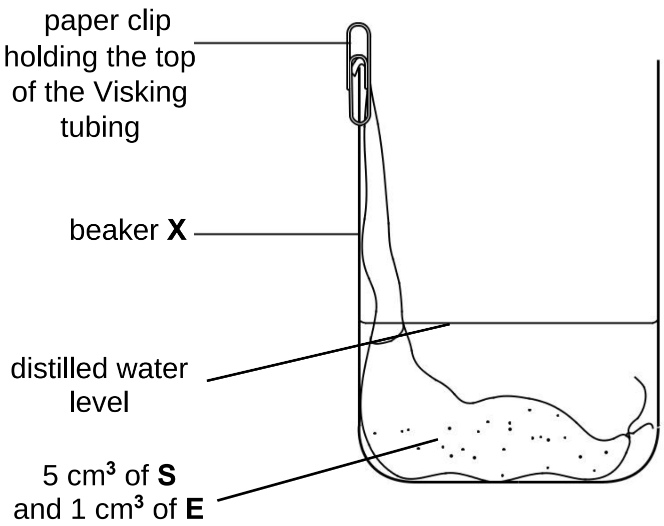


Fig. 1.1

Proceed as follows:

1. Tie a knot in the Visking tubing as close as possible to one end, so that it seals the end.

2. To open the other end, wet the Visking tubing and rub the tubing gently between your fingers.

3. Put 5 cm^3 of **S** into the Visking tubing.

4. Put 1 cm^3 of **E** into the Visking tubing.

5. Rinse the outside of the Visking tubing by dipping it into the water in the container labelled **V**.

Look carefully at Fig. 1.1. This has been set up so that the volume of water is as small as possible to cover the Visking tubing. The part of the Visking tubing containing the mixture is on the bottom of the beaker.

6. Put the Visking tubing into the beaker, labelled **X**, as shown in Fig. 1.1.

7. Put **W** into the beaker up to the level shown on Fig. 1.1 using a syringe so that you can measure the volume of **W**.

(a) **(i)** State the volume of **W** needed to reach the water level as shown in Fig. 1.1.

volume of **W** = ... cm^3 [1]

a) Appropriate volume of water used

8. Leave the apparatus for 15 minutes.

While you are waiting, continue with Question 1.

9. After 15 minutes, remove the Visking tubing and put it into the container labelled '**For waste**'.

You are required to:

- prepare a serial dilution of the 1.0% reducing sugar solution, **R1**
- carry out the Benedict's test for the known concentrations of reducing sugar and the water surrounding the Visking tubing
- use the results to estimate the concentration of reducing sugar in the water surrounding the Visking tubing.

You are provided with:

labelled	contents	hazard	volume / cm^3
W	distilled water	none	250
R1	1.0% reducing sugar solution	none	30
Benedict's	Benedict's solution	irritant	30

It is recommended that you wear suitable eye protection.

If **Benedict's** comes into contact with your skin, wash it off immediately with cold water.

(ii) You are required to make a **serial** dilution of the 1.0% reducing sugar solution, **R1**, which reduces the concentration of reducing sugar solution by **half** between each successive dilution. You will also need to set up a control, **C**.

You are required to make up at least $10 cm^3$ of each concentration of reducing sugar solution in the small glass vials provided.

Complete Table 1.2 to show how you will make the concentrations of the reducing sugar solutions, **R2, R3, R4 and R5**, and show how you will set up the control, **C**.

Table 1.2

	R1	R2	R3	R4	R5
Concentration of reducing sugar solution / %	1.00	0.50	0.25	0.125	0.0625
Label of reducing sugar solution to be diluted		R1	R2	R3	R4
Volume of reducing sugar solution to be diluted/ cm^3		10.0	10.0	10.0	10.0
Volume of distilled water, W, to make the dilution/ cm^3		10.0	10.0	10.0	10.0
Description of the control, C: **a) Using a syringe, transfer 5.0 cm^3 of distilled water (into a small container) instead of 5.0 cm^3 of glucose solution;;**					

[4]

b) correct precision for reducing sugar concentration;
c) correct precision for both volumes of reducing sugar and distilled water (1 d.p.);
d) equal volume of reducing sugar and distilled water;
e) minimal volume after transfer is 10.0 cm^3;
f) correct calculation of concentration of reducing sugar;
g) correct label of reducing sugar solution to be diluted;

10. Set up a boiling water-bath ready for step 12.

11. Prepare the concentrations of reducing sugar solution, as decided in **(a)(ii)**, in the glass vials provided.

12. Carry out the Benedict's test on each of the concentrations of reducing sugar solution and record your results in **(a)(iii)**.

    You will need to use $2 cm^3$ of each of the concentrations of reducing sugar solution with $3 cm^3$ of Benedict's solution.

13. Test each solution **separately** and record in **(a)(iii)** the time taken for the **first** appearance of any colour change. If there is no colour change after 180 seconds record as 'more than 180'.

(iii) Prepare the space below and record your results.

Table of time taken for time taken for first appearance of any colour change / s for different reducing sugar concentrations / %

Concentration of reducing sugar solution / %	time taken for first appearance of any colour change / s
0.0000 (control)	more than 180
0.0625	
0.1250	
0.2500	
0.5000	
1.0000	

a) MMO – as long as students can produce the table of results;
b) Table title with units;
c) correct headings with units;
d) records time for at least four concentrations of reducing sugar;
e) correct precision of time (whole numbers);;
f) trend (shortest time taken for 1st appearance for 1% reducing sugar solution → longest time taken for 1st appearance for 0.0625% reducing sugar solution);;
g) control concentration with time taken (more than 180);;

[5]

(iv) Describe **one** significant source of error when carrying out steps 12 and 13. [1]

a) appropriate error with reason, e.g. colour change (can be subjective), difficult to judge first appearance of colour change;;

R: other sources of error NOT from steps 12 and 13 (e.g. having more concentrations of reducing sugar solution → step 11)

Before proceeding, check that you have carried out step 9 on page 3.

14. Carry out the Benedict's test on a sample taken from beaker **X** (sample **X**) and record the time taken for the first colour change to appear.

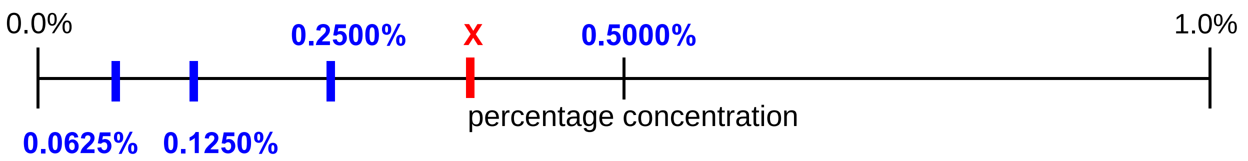
(v) State the time taken for the first colour change for sample **X**.

time taken = ... [1]

a) records time taken for first colour change for X in seconds (whole numbers);;

(vi) Complete Fig. 1.3 to show:

- the positions of each of the percentage concentrations of reducing sugar solution
- an estimate of the concentration of reducing sugar in the sample **X**, using a letter **X**.



[2]

Fig. 1.3

a) correctly labels Fig.1.3 with reducing sugar concentrations;;
b) correctly places X on Fig.1.3 in the correct position according to table of results and time taken for X in (v);;

(vii) Describe how you could use this procedure to produce a more accurate estimate of the concentration of reducing sugar in the sample **X** than the one given in **(a)(vi)**.
Do **not** include the use of a colorimeter in your answer. [3]

a) increase number of concentrations of reducing sugar concentrations or examples of concentrations;;
b) uses proportional / simple dilution or serial dilution to make concentrations;;
c) reference to drawing a graph and reading off estimate for the concentration of reducing sugar in sample X;;
d) comparing times for sample X with times for known concentrations of reducing sugar / carry out at least 3 replicates;;

Max 3

(b) A scientist investigated the effect of temperature on the activity of the enzyme in the Visking tubing.

All other variables were kept constant.

The quantity of reducing sugar diffusing through the wall of the Visking tubing was measured by a dye in the surrounding solution. The dye reacted with the reducing sugar. The more reducing sugar present the more intense the colour.

A colorimeter was used to measure the absorbance of light by the coloured solution. The absorbance of light by pure water is 0.00 arbitrary units.

The results are shown in Table 1.4.

Table 1.4

temperature / °C	absorbance of light by the coloured solution / arbitrary units
30	0.90
41	1.46
49	1.58
59	1.10
70	0.65

Use a sharp pencil for graphs.

(i) Plot a graph of the data shown in Table 1.4.

[4]

a) appropriate title given with units;
b) appropriate size i.e. at least ¾ of grid;
c) correct choice of axes AND units, (*x*-axis) temperature / °C + (*y*-axis) absorbance of light by the coloured solution / arbitrary units;
d) intervals of the graph are equidistant AND no awkward scale, scale on *x*-axis: 10 to 2 cm, labelled at least each 2 cm + origin labelled 30 + scale on *y*-axis: 0.2 to 2 cm, labelled at least each 2 cm;
e) correct plotting of 5 points as a small cross or dot in circle, to within half a small square;
f) appropriate line of best fit (5 plots + ruled sharp lines/curve exactly point to point);
g) and smooth line (in less than 1mm thickness);
h) no extrapolation beyond extreme measured data;

(ii) Explain the difference in the absorbance of light between 49 °C and 70 °C. [3]

a) As temperature increases to 49 °C, increase in kinetic energy of enzyme and substrate molecules;
b) Resulting in increased frequency of successful collisions / more enzyme substrate complexes formed per unit time;
c) At 70 °C, temperature is beyond optimum, excessive molecular motion will disrupt/break the weak bonds / (hydrogen bonds, hydrophobic interactions, ionic bonds);
d) resulting in lost of 3D conformation of enzyme;
e) enzyme active site no longer complementary to substrate;
f) substrate unable to bind to enzyme at active site;
g) fewer enzyme substrate complexes formed per unit time;
h) enzyme denatures;

Max 3

[Total: 24]

2 **L1** is a slide of a stained transverse section through a plant leaf.

You are not expected to be familiar with this specimen.

You are required to:

- use the eyepiece graticule to measure part of a leaf and a vascular bundle
- use these measurements to calculate the depth of the vascular bundle as a percentage of the depth of the leaf
- draw a plan diagram of part of the leaf.

(a) The eyepiece graticule in the microscope can be used to measure different tissues. Select a part of the leaf on **L1** which shows the widest part of the leaf (mid-rib) shown by **Y** in Fig. 2.1.

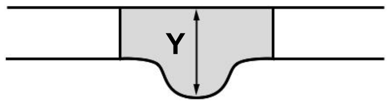


Fig. 2.1

(i) Use the eyepiece graticule in the microscope to measure:

- the depth of the leaf at **Y**
- the depth of the vascular bundle at **Y**.

depth of leaf .... eyepiece graticule units

depth of vascular bundle .... eyepiece graticule units [1]

a) records measurements of the depth of the leaf AND the depth of the vascular bundle (view at 10x objective lens magnification)

R: depth of leaf ≤ depth of vascular bundle
R: depth of leaf or depth of vascular bundle > 100 eyepiece graticule units (due to viewing at 40x objective lens magnification)

(ii) Use the measurements from **(a)(i)** to calculate the depth of the vascular bundle as a percentage of the depth of the leaf.

You may lose marks if you do not show your working.

Depth of vascular bundle / depth of leaf x 100%

= .... %

a) shows measurement of the vascular bundle divided by the measurement of the depth of the leaf, multiplied by 100;;
b) round off to 3 s.f.

answer = ...% [2]

Use a sharp pencil for drawing.

(iii) Use the measurements from **(a)(i)** to help you draw a large plan diagram of the section of the leaf shown by the shaded area in Fig. 2.2.

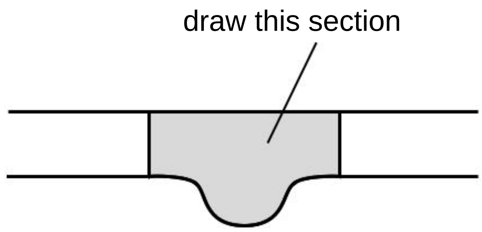
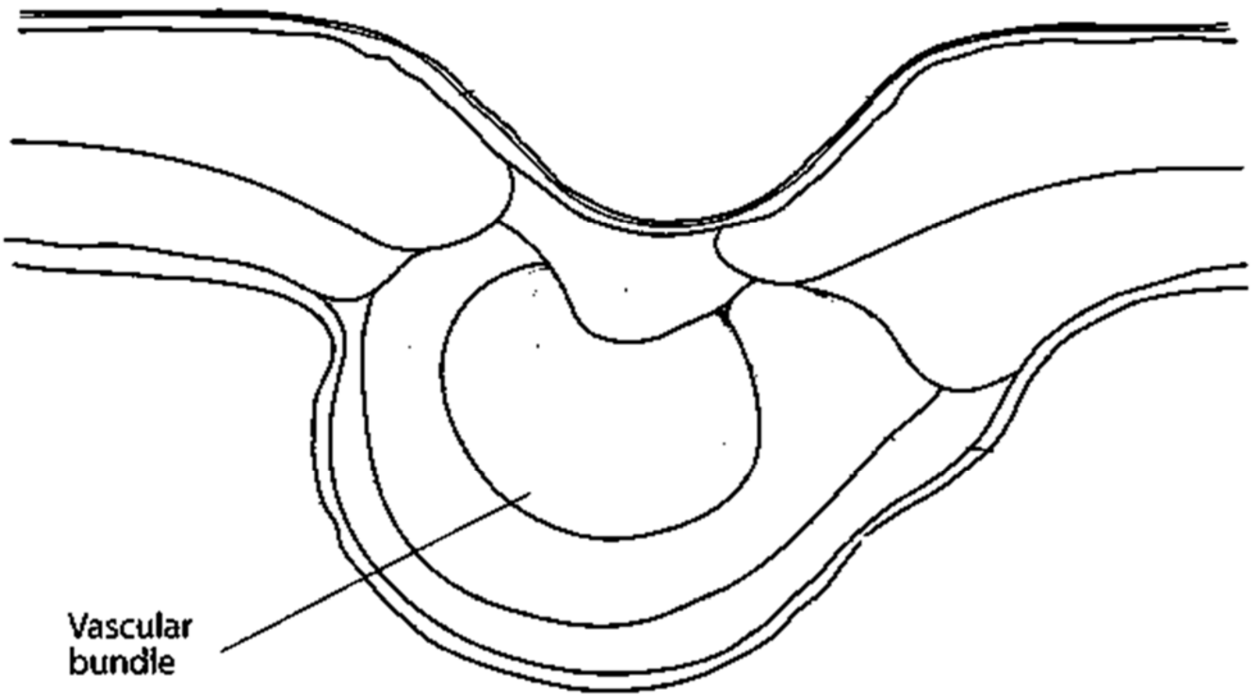


Fig. 2.2

You are expected to draw the correct shape and proportions of the different tissues.

Use **one** ruled label line and label to identify the vascular bundle.



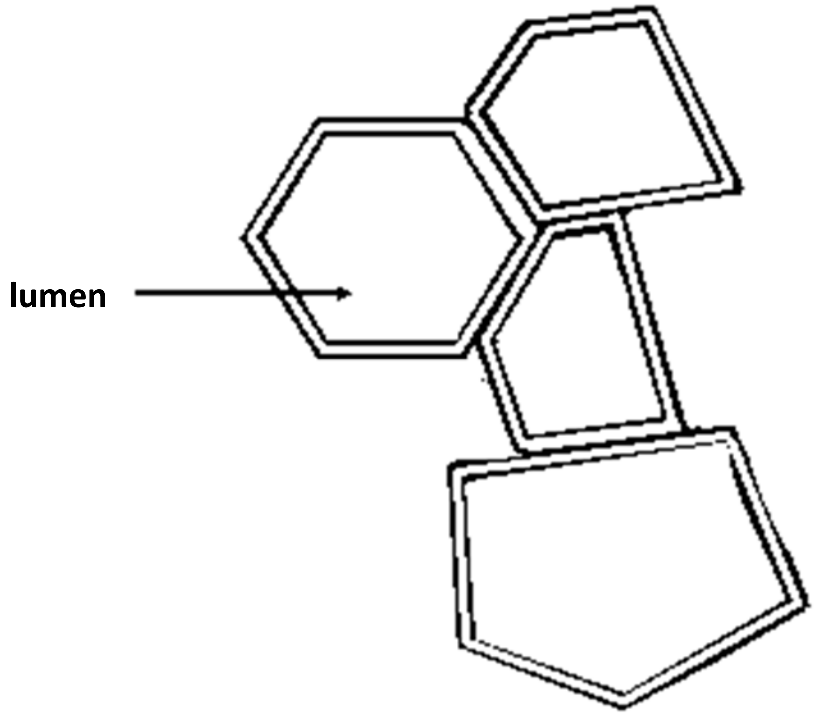
a) No shading;
b) minimum size at least 90 mm;
c) no individual cells;
d) at least 4 lines drawn;
e) draws correct section of leaf;;
f) draws correct proportion of the vascular bundle in relation to the depth of the leaf;
g) correct shape of the vascular bundle;
h) uses one label line + one label to the vascular bundle;;

[5]

(iv) Observe the vascular bundle in the central part of the leaf on **L1**.
Select one group of **four** adjacent (touching) xylem vessel elements. Each element must touch at least one of the other elements.

Make a large drawing of this group of **four** xylem vessel elements.

Use **one** ruled label line and label to identify the lumen of **one** xylem vessel element.



a) **quality of the line for the outer wall of vessel elements (thin line);**
b) **minimum size of at least 40 mm across the largest vessel element;**
c) **only four vessel elements drawn, each touching at least one of the other vessel elements;;**
d) **walls of vessel elements drawn as two lines;;**
e) **at least one vessel element drawn with more than four sides;;**
f) **uses one label line + one label to lumen;;**

[5]

(b) Fig. 2.3 is a photomicrograph of a stained transverse section through a different type of leaf. You are not expected to be familiar with this specimen.

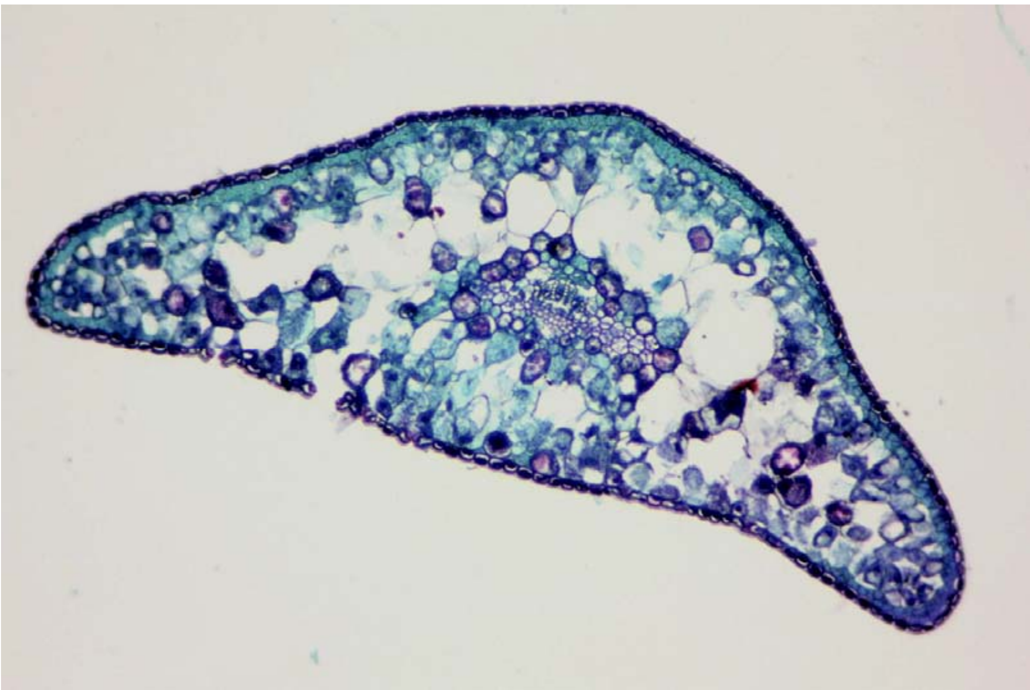


Fig. 2.3

Prepare the space below so that it is suitable for you to record the observable differences between the leaf sections on **L1** and in Fig. 2.3.

Record your observations in the space you have prepared.

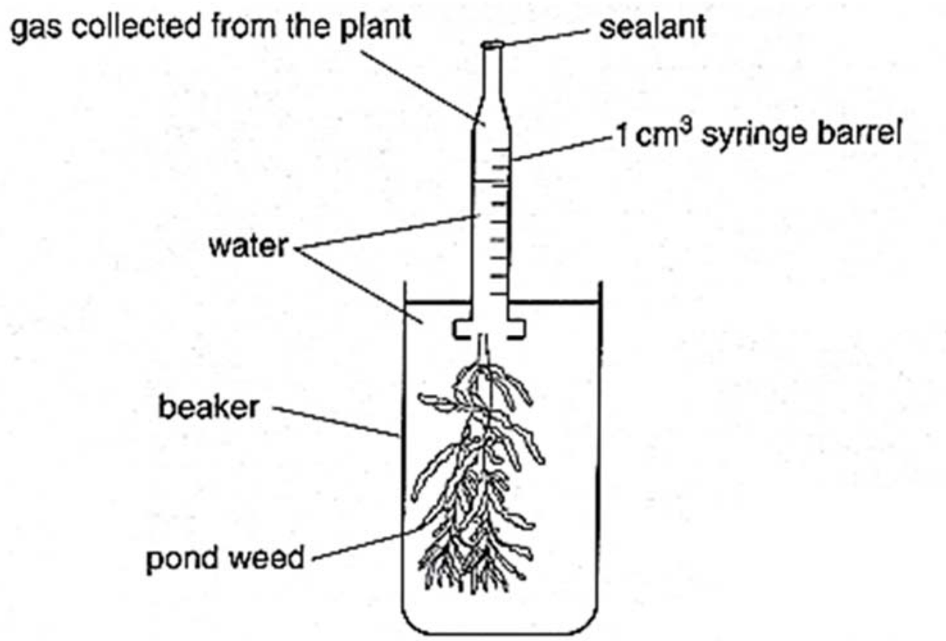
a) PDO - organises comparison into three columns with one column for features, one column headed L1 and one column headed Fig. 2.3;;
b) Any 3 differences (Max 3)

Feature	L1	Fig. 2.3
Differentiation of mesophyll cells OR Presence of palisade mesophyll layer	Mesophyll cells differentiated into palisade and spongy mesophyll cells Palisade mesophyll layer present;	Mesophyll cells not differentiated into palisade and spongy mesophyll cells / remain undifferentiated Palisade mesophyll layer absent;
Density of mesophyll cells	More dense;	Less dense;
Size/proportion of vascular bundle compared to depth of leaf	Vascular bundle takes a larger proportion of the leaf;	Vascular bundle takes a smaller proportion of the leaf;
Number/size of air spaces	Less/smaller air spaces;	More/larger air spaces;

[4]

[Total: 17]

3 The rate of photosynthesis can either be measured by the rate at which carbon dioxide is taken in or the amount of oxygen that is given out. Some water plants release bubbles of gas from a freshly cut stem when illuminated. Light intensity is controlled using five filters, F1, F2, F3, F4, F5.



Different water plants are adapted to different light intensities. A sun-loving water plant is adapted to high light intensities while a shade-loving water plant is adapted to low light intensities.

Using this information, the set-up above and your own knowledge, design an experiment to investigate the effect of light intensity on photosynthesis in sun and shade plants.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- Sun plant
- Shade plant
- Bench lamp with 60 W bulb
- 5 filters (F1, F2, F3, F4, F5) which can be adjusted to allow different amounts of light to pass through
- 1% sodium hydrogencarbonate solution

You may select from the following apparatus and use appropriate additional apparatus:

- Normal laboratory glassware, e.g. a variety of different sized beakers, measuring cylinders, and syringes for measuring volumes
- Forceps
- Timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is repeatable by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

Marking scheme

Compulsory (5 marks total)

a)	Broad outline -1m A**T**H**P**DR	**General idea** *How would you measure the dependent variable?* Under **T**heory measuring oxygen evolved (via photolysis) in response to different light intensities over a set period of time;; OR Under **P**rocedure Mark will be awarded if the broad outline of experiment is reflected in the procedure.
b)	Independent variable -1m ATH**P**DR A**N**CR D**I**CE	**State what the independent variable is, use at least five different values with equal intervals, good range, with units** Under **P**rocedure (within **N**umbered steps section) **I**ndependent variable: Light intensity (20%, 40%, 60%, 80%, 100% transmission) A! Highest low limit: 30% Lowest high limit: 90% A! Lux R! Au

<table>
<tr>
<td>c)</td>
<td>Method to vary independent variable -1m</td>
<td>
Plan suitable method to vary the independent variable.

<u>Under Procedure</u>
Place filters with the different light transmissions in front of the lamp.
</td>
</tr>
<tr>
<td>d)</td>
<td>Table -1m

ATHPDR</td>
<td>
Shows how results are to be presented in the form of a table with independent and dependent variables in appropriate columns / rows. Units must be correct.

<u>Under Data Recording</u>

<table>
<tr>
<th rowspan="2">Type of plant</th>
<th rowspan="2"><u>Light intensity / %</u></th>
<th colspan="4">Distance moved / Volume of oxygen evolved in 10 min/cm^3</th>
<th rowspan="2"><u>Average</u> rate of photosynthesis/ Average rate of oxygen evolved / cm^3 s^{-1}</th>
</tr>
<tr>
<th>replicate 1</th>
<th>replicate 2</th>
<th>replicate 3</th>
<th>Average</th>
</tr>
<tr><td rowspan="5">Sun plant</td><td>20</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>40</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>60</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>80</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>100</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="5">Shade plant</td><td>20</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>40</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>60</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>80</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>100</td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

A! Separate table for sun and shade plant
No need control
Units: To follow raw data collected
</td>
</tr>
<tr>
<td>e)</td>
<td>Risk/ safety -1m

ATHPDR</td>
<td>
Risk / safety (tabulate and use risk assessment unique to the experiment)

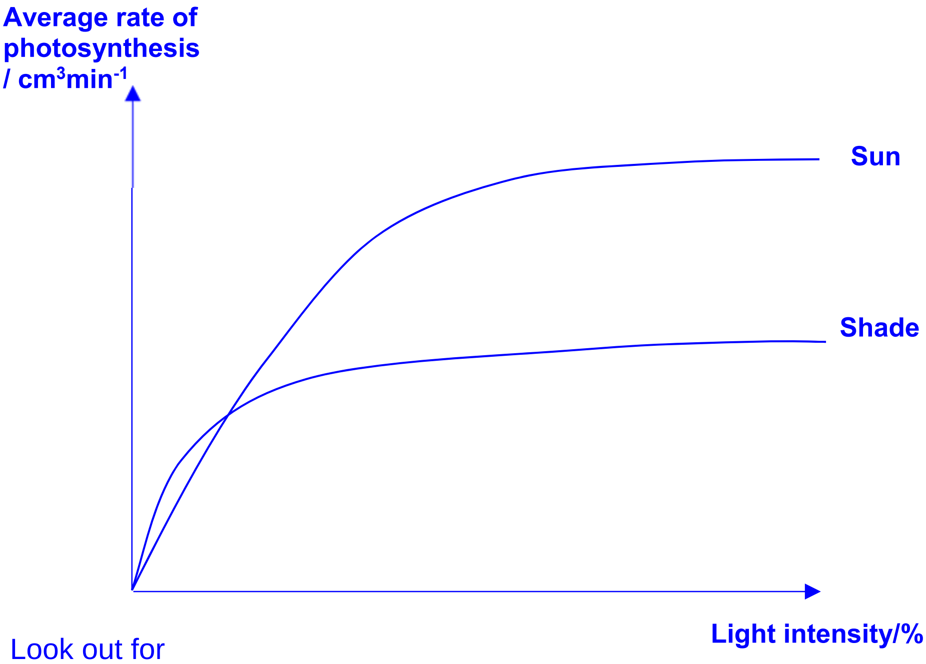
<u>Under Risk Assessment (Any 2, 0.5m each)</u>

<table>
<tr><th>Safety hazard</th><th>Precaution</th></tr>
<tr><td>Cutting plants with sharp scalpel</td><td>Use forceps to hold the plant to prevent cutting of fingers;</td></tr>
<tr><td>Sodium hydrogen carbonate is an irritant;</td><td>Wear safety gloves when handling the chemical;</td></tr>
<tr><td>Risk of electrocution when handling the electrical bench lamp with wet hands</td><td>Avoid touching electrical socket with wet hands / keep hands dry when handling the lamp;</td></tr>
<tr><td>Bulb of lamp will get hot when used</td><td>Avoid touching the bulb when lamp is in use;</td></tr>
</table>
</td>
</tr>
</table>

Max 9m from any of the below points

f)	Method + Scientific reasoning -1m ATHPDR	**Plan a suitable method that involves monitoring/measuring the dependent variable in response to the varied independent variable over a period of time/ set interval** *If experiment is fundamentally wrong, do not award for this mark.* Under Theory *How would the independent variable affect the dependent variable?* An increase light intensity will increase the rate of photophosphorylation as more electrons are excited. In order to fill the electron gap, the rate of photolysis of water increases, where water is broken down to form of H^+ and oxygen, which is given off. The higher the light intensity, the higher the rate of oxygen formation;; (The sun plant would have higher rate of photosynthesis at high light intensities while shade plants would have a higher rate of photosynthesis at low light intensities.)
g)	Dependent variable -1m ATHPDR ANCR DICE	**State what is the dependent variable.** Under Procedure (within Numbered steps section) Dependent variable: (Rate of photosynthesis measured by) volume of oxygen evolved / distance of meniscus moved;
h)	Method -1m How to measure / monitor dependent variable	**Specifies method of measuring / monitoring dependent variable** Under Procedure Record the change in liquid level / volume of gas over a period of (10 min);
i)	Controlled variables to improve accuracy or reliability -1m ATHPDR ANCR DICE	**Identifies at least two variables to control**; (Mark if specified in diagram) Under Procedure (within Numbered steps section) Controlled variable (Any 2, 0.5m each) 1. Distance of lamp from plant; 2. Distance of filter from lamp; 3. Length of plant / number of leaves / number of plants / size of leaves; 4. Duration of exposure to light; 5. Volume of sodium hydrogen carbonate added to the water; 6. Temperature; R! Concentration of sodium hydrogen carbonate (Given in question)
j/k)	Controlled variables -2m ATHPDR ANCR DICE	**Describes how two identified variables are controlled. (Must specify appropriate value)** Under Procedure (within Numbered steps section) Controlled variable (based on the two identified variables, 1m each) 1. Place a lamp of 60 W bulb at a distance of 15 cm away from the beaker of water;; 2. Cut a branch of plant 2 cm long and place it in a glass filter funnel;; 3. Turn on the lamp for a period of 10 min;; 4. Add 2 cm^3 of sodium hydrogencarbonate added;;

		5. Use a thermostatically-controlled water bath;;
l)	Control -1m ATH**P**DR AN**C**R	**Describe control** <u>Under **P**rocedure</u> Conduct a **c**ontrol or * experiment using boiled and cooled sun and shade plant / no plant / plant without leaves;; (*with the same set up / experimental conditions.) (This is to ensure that the changes in volume of oxygen is due to the change of rate of photosynthesis.) R! Zero light intensity is not a good control because gas will still be produced. (Specific to photosynthesis)
m/n)	Method -2m ATH**P**DR A**N**CR	**Steps to take that would ensure the validity of the experimental results. (Not the same as controlled variable)** <u>Under **P**rocedure</u> (within **N**umbered steps section) (any 2, 1m each) 1. Adding excess sodium hydrogen carbonate into the water (to ensure sufficient concentration of CO_2);; 2. Use a fresh solution of sodium hydrogencarbonate for each replicate (to ensure sufficient concentration of CO_2);; 3. Conduct experiment in a dark room / eliminate all other light sources / ensure all other light sources are constant (to prevent heating effect);; 4. Use a cool light source / water screen in front of the lamp / use thermostatically-controlled water bath (to prevent heating effect of lamp)**unless about thermostatically controlled water bath*;; 5. Ensure the water level of thermostatically-controlled water bath is above the water level in the beaker (to ensure homogenous temperature of liquid);; 6. Pick actively bubbling plants (for observable displacement of water) 7. Ensure sufficient number of leaves (for observable displacement of water);; 8. AVP;;
o)	Method -1m ATH**P**DR A**N**CR DIC**E**	**Plan a method for equilibration** <u>Under **P**rocedure</u> (within **N**umbered steps section) Immerse the plant in the sodium hydrogen carbonate solution and illuminate it for fixed time (10 minutes) for **e**quilibration before starting to collect oxygen;;
p)	Reliability – 1m ATH**P**DR ANC**R**	**Reference to doing two more replicates** <u>Under **P**rocedure</u> (within **N**umbered steps section) Perform the experiment in sets of 3.
q)	Reproducibility -1m ATH**P**DR ANC**R**	**Reference to repeating at least two more time with different experimental subjects.** <u>Under **P**rocedure</u> (within **N**umbered steps section) **R**epeat experiment for another 2 times using another branch / plant, to ensure reproducibility of data;;

r)	Accuracy -1m ATHPDR ANCR	**Repeating experiment with smaller intervals** Under Procedure (within Numbered steps section) Repeating the experiment with filters of smaller interval of light transmission at 10% intervals / any suitable interval;
s)	Graph -1m ATHPDR	**Axes drawn must be correct, two graphs on the same axes**  Look out for • Shape: Shade > Sun at low light intensities, Sun > Shade at high light intensities, Plateau on both graphs • A! Rate of photosynthesis and light intensity